ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	147302
ToLID	**aRanIta1**
Species	Rana italica
Class	Amphibia
Order	Anura

Genome Traits	Expected	Observed
Haploid size (bp)	5,101,732,283	5,155,001,779
Haploid Number	13 (source: direct)	13
Ploidy	3 (source: ancestor)	2
Sample Sex	unknown	unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for hap1: 6.8.Q64
Obtained EBP quality metric for hap2: 6.8.Q64

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Observed Ploidy is different from Expected

. BUSCO single copy value is less than 90% for hap2

Curator notes

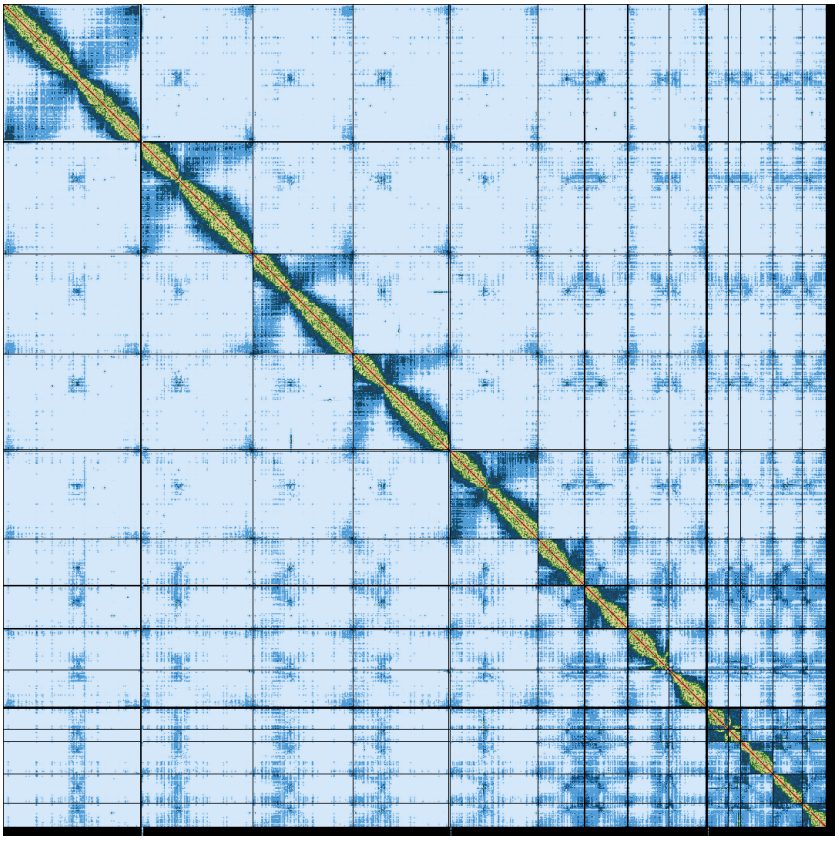
. Interventions/Gb: 20
. Contamination notes: "FCS-GX did not detected contaminants in the curated assemblies. Our adapted blobplots will just show any contaminants when they are detected otherwise they are empty."
. Other observations: "First, Hifiasm assemblies were scaffolded with yahs without allowing for contig breaks. Then it was curated using the diploid map to produce the curated haplotypes under review: Curation made 15 cuts in contigs, 71 breaks at gaps and 118 joins. Finally, the mitogenome was assembled into a single circular contig 22,497 bp long that will be uploaded to ENA together with the reviewed assembly. Note that the links provided point to a shared folder with the High Resolution maps (HR) allowing multimappings (mq0) or not (mq10). We have done a final round of curartion for each hap. Please start your review of from the aRanIta1_hap*_mq10.extensions.HR.pretext.savestate_1 of each hap."

Quality metrics table

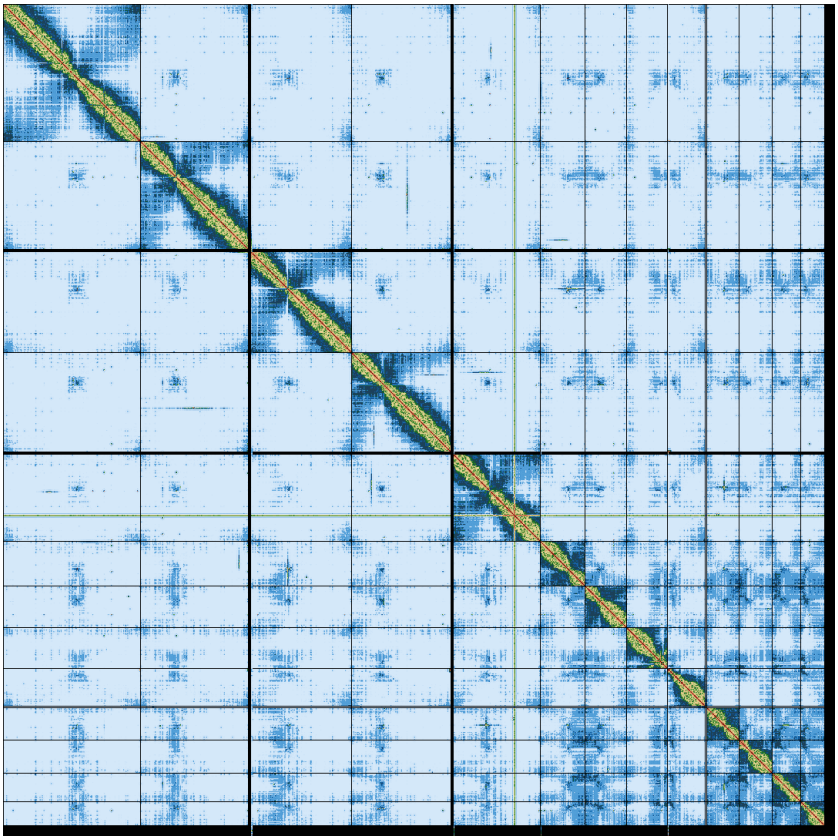
Metrics	Pre-curation hap1	Pre-curation hap2	Curated hap1	Curated hap2
Total bp	5,149,033,152	5,028,367,072	5,155,001,779	5,022,407,845
GC %	44.56	44.5	44.56	44.5
Gaps/Gbp	502.42	532.78	508.83	535.6
Total gap bp	517,400	535,800	524,600	538,000
Scaffolds	344	330	331	311
Scaffold N50	676,956,069	596,098,675	593,253,258	600,721,939
Scaffold L50	3	4	4	4
Scaffold L90	11	11	11	11
Contigs	2,931	3,009	2,954	3,001
Contig N50	5,393,737	4,359,970	5,334,790	4,359,970
Contig L50	279	317	283	318
Contig L90	1,186	1,351	1,202	1,349
QV	64.094	64.3927	64.0865	64.4013
Kmer compl.	94.8376	94.0275	94.8402	93.9407
BUSCO sing.	90.1%	87.5%	90.0%	87.5%
BUSCO dupl.	2.9%	2.7%	2.9%	2.7%
BUSCO frag.	1.9%	2.3%	2.0%	2.3%
BUSCO miss.	5.1%	7.5%	5.1%	7.5%

BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: vertebrata_odb12 (genomes:119, BUSCOs:3390)

HiC contact map of curated assembly

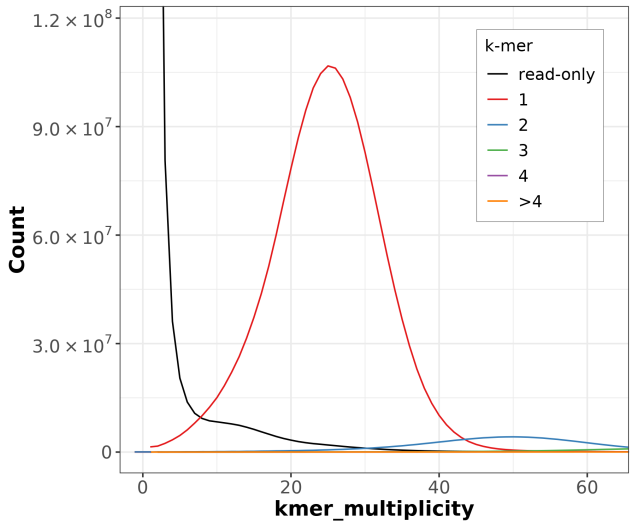


hap1 [LINK]

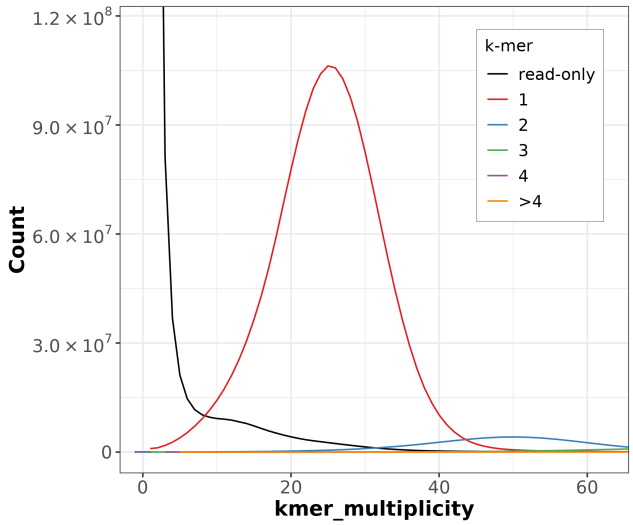


hap2 [LINK]

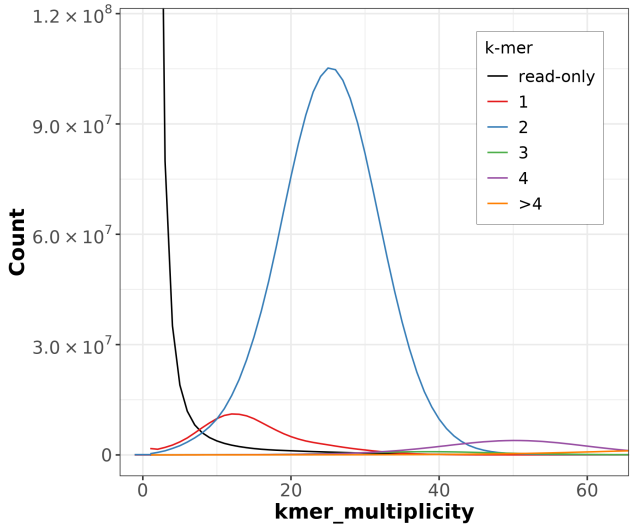
K-mer spectra of curated assembly



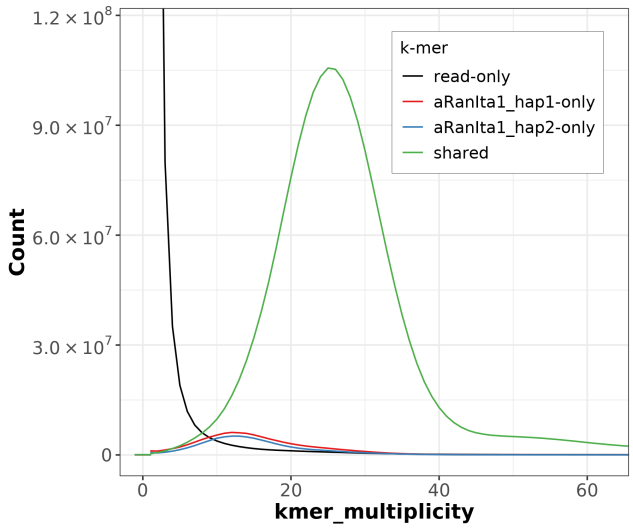
Distribution of k-mer counts per copy numbers found in **aRanIta1_hap1** (hapl.)



Distribution of k-mer counts per copy numbers found in **aRanIta1_hap2** (hapl.)

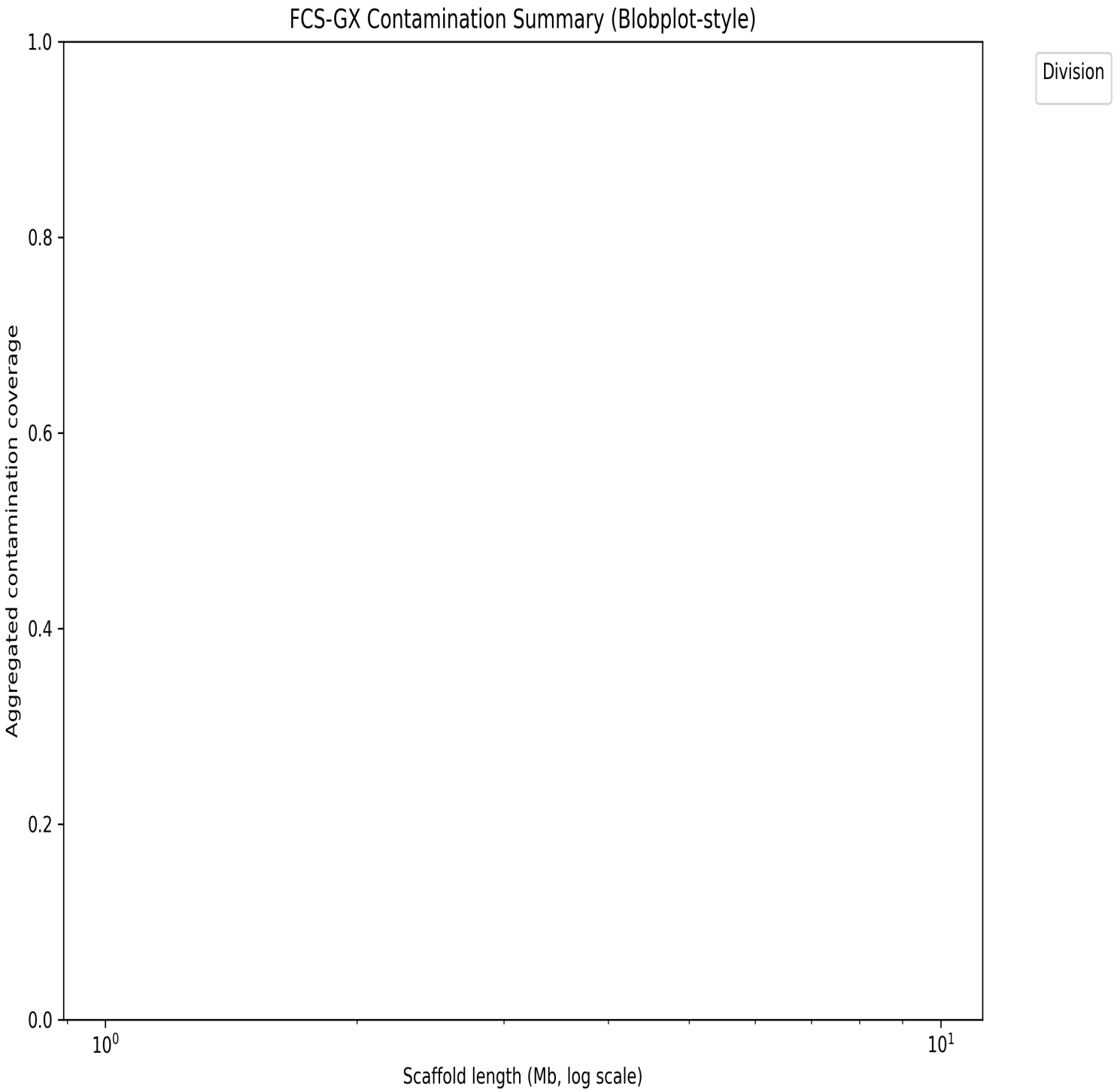


Distribution of k-mer counts per copy numbers found in asm (dipl.)

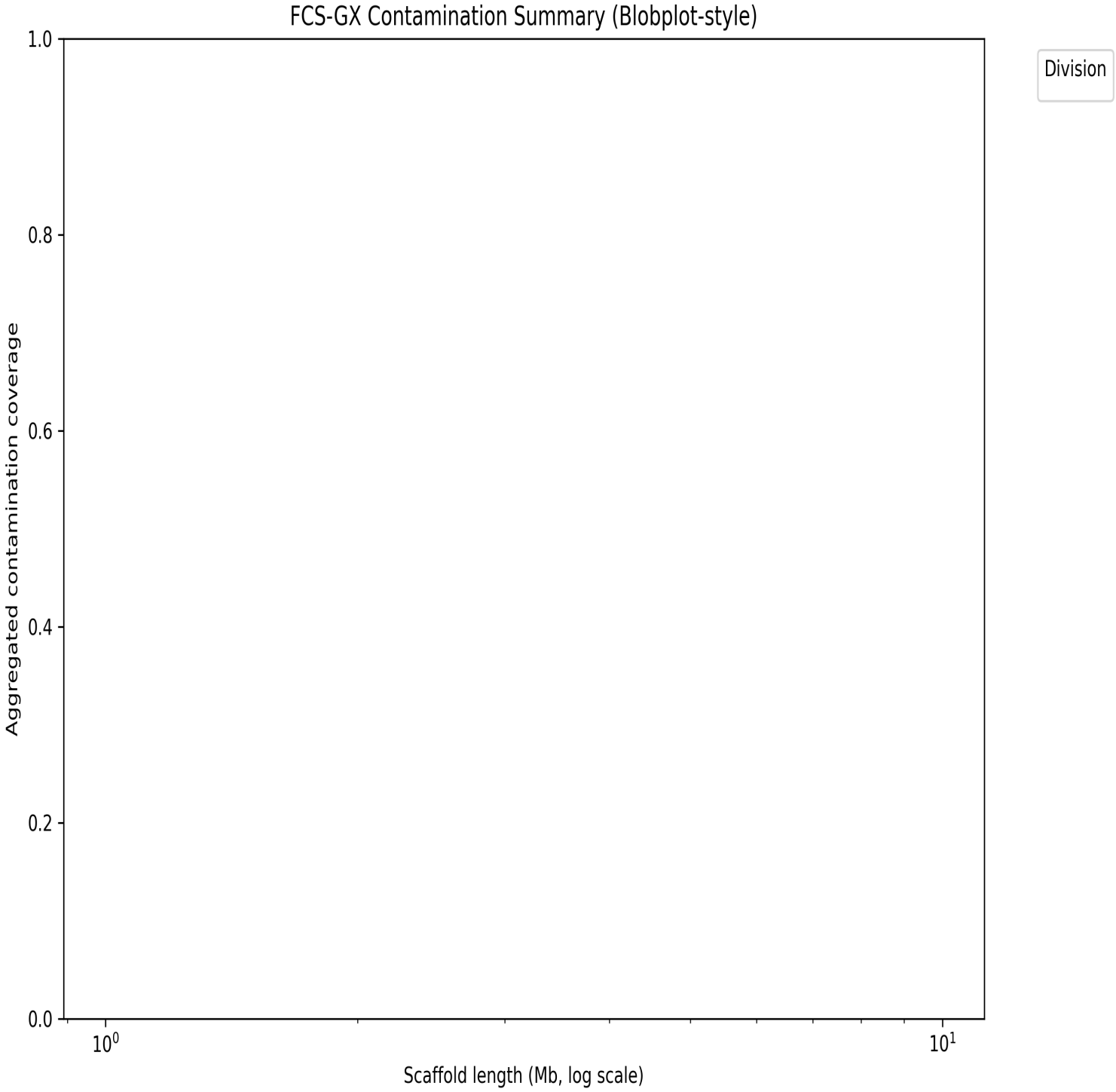


Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening



hap1. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.



hap2. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	HiFi	Hi-C
Coverage	27x	240x

Assembly pipeline

- **CLAWS pipeline**
    |_ *ver:* 3.1
    |_ *key param:* NA
- **Hifiasm**
    |_ *ver:* 0.24.0
    |_ *key param:* --telo-m TTAGGG
- **YaHS**
    |_ *ver:* 1.2a
    |_ *key param:* --no-contig-ec
- **Merqury**
    |_ *ver:* 1.3
    |_ *key param:* NA
- **BUSCO**
    |_ *ver:* 6.0.0
    |_ *key param:* NA
- **tidk**
    |_ *ver:* 0.2.65
    |_ *key param:* NA
- **chromap**
    |_ *ver:* 0.3.2
    |_ *key param:* NA
- **FOAM pipeline**
    |_ *ver:* 0.5
    |_ *key param:* NA
- **mitos**
    |_ *ver:* 2.1.3
    |_ *key param:* NA

Curation pipeline

- **Blobtoolkit DToL Nextflow pipeline**
    |_ *ver:* 0.6.0
    |_ *key param:* NA
- **FCS-GX**
    |_ *ver:* 0.5.5
    |_ *key param:* NA
- **PretextViewAI**
    |_ *ver:* 1.0.5
    |_ *key param:* NA
- **SANGER tola-agp-tpf-utils**
    |_ *ver:* 1.2.1
    |_ *key param:* NA

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